



Bank Marketing Term Purchase Prediction

Overview

The data is related to direct marketing campaigns (phone calls) of a Portuguese banking institution. The classification goal is to predict if the client will subscribe to a term deposit (variable y).

Goal

Predict whether a customer will subscribe to a term deposit based on demographic, financial, and campaign interaction data.

Compare multiple ML models and evaluate performance.

To have an comprehensive understanding of client's information and there behaviour

To Understand about the Profitability and Risk of allotting a term deposit

Variables Information

Input variables:

bank client data:

1 - age (numeric)

2 - job : type of job (categorical:

"admin.", "unknown", "unemployed", "management", "housemaid", "entrepreneur", "student",
"blue-collar", "self-employed", "retired", "technician", "services")

3 - marital : marital status (categorical: "married", "divorced", "single"; note: "divorced" means divorced or widowed)

4 - education (categorical: "unknown", "secondary", "primary", "tertiary")

5 - default: has credit in default? (binary: "yes", "no")

6 - balance: average yearly balance, in euros (numeric)

7 - housing: has housing loan? (binary: "yes","no")

8 - loan: has personal loan? (binary: "yes","no")

related with the last contact of the current campaign:

9 - contact: contact communication type (categorical: "unknown","telephone","cellular")

10 - day: last contact day of the month (numeric)

11 - month: last contact month of year (categorical: "jan", "feb", "mar", ..., "nov", "dec")

12 - duration: last contact duration, in seconds (numeric)

other attributes:

13 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)

14 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)

15 - previous: number of contacts performed before this campaign and for this client (numeric)

16 - poutcome: outcome of the previous marketing campaign (categorical: "unknown","other","failure","success")

Output variable (desired target):

17 - y - has the client subscribed a term deposit? (binary: "yes","no")

Data Cleaning

Missing Value Analysis

The dataset contains missing values across several features. These were handled based on both percentage of missing data and business reasoning.

1 Job Type

Missing values: 288 (~0.6%)



Reason: Customers may intentionally choose not to disclose their job.

Imputed the Empty Rows as Unknown Category

2 Education

Missing values: 1857 (~4.1%)

Reason: Customers may avoid sharing education details.

Imputed the Empty Rows as Unknown Category

3 Contact

Missing values: 13,020 (~28.8%)

Observation: Missing values are already labeled as "unknown"

Retained "unknown" as a valid category instead of imputing.

4 Previous Outcome (poutcome)

Missing values: 36,959 (~81.7%)

Extremely high missing percentage (>70%)

Can introduce data leakage (contains past campaign outcomes)

This column was dropped due to:

Low data reliability

Risk of misleading the model

Data Leakage Handling

Duration Feature

Description: Duration of the last call with the customer

This value is only known after the call happens

Not available at prediction time

The duration column was removed to prevent data leakage and ensure realistic model performance.

Class Imbalance

Majority class: **Non-subscribers (~88%)**

Minority class: **Subscribers (~12%)**

This indicates a **highly imbalanced dataset**.

Used techniques such as:

class weighting / sample weighting

precision-recall based evaluation

threshold tuning

Exploratory Data Analysis

Univariate Analysis

Box Plot (For Numerical Features) ('age', 'balance', 'campaign', 'pdays', 'previous')

Age

- Both subscribed (Yes) and non-subscribed (No) groups show outliers.
- Customers who subscribed tend to be slightly older compared to those who did not.

Balance

- Outliers are present in both groups.
- Interestingly, non-subscribed customers have higher account balances on average than subscribed customers.

Campaign (Number of Contacts)

- Customers who did not subscribe were contacted more frequently.
- The distribution is more spread out, indicating higher marketing effort on non-converting customers.

Pdays (Days Since Last Contact)

- Customers who subscribed were contacted after a longer gap.
- This suggests that **less frequent but well-timed

Bar Chart (For Categorical Features) ('job', 'marital', 'education', 'default', 'housing', 'loan', 'contact')

Job

- Customers in management roles have the highest subscription rates.
- Blue-collar customers have the lowest subscription rates, indicating weaker conversion in this segment.

Marital Status

- All categories (single, married, divorced) show similar proportions of subscribers and non-subscribers.
- Marital status does not significantly impact subscription behavior.

Education

- Subscription rates are fairly consistent across education levels.
- Education does not appear to be a strong influencing factor.

Default (Credit Default Status)

- Customers who have defaulted on loans did not subscribe to term deposits.
- This indicates that financially risky customers are less likely to invest.

Housing Loan

- Customers with no housing loan are more likely to subscribe.
- Those with housing loans show lower subscription rates, likely due to financial obligations.

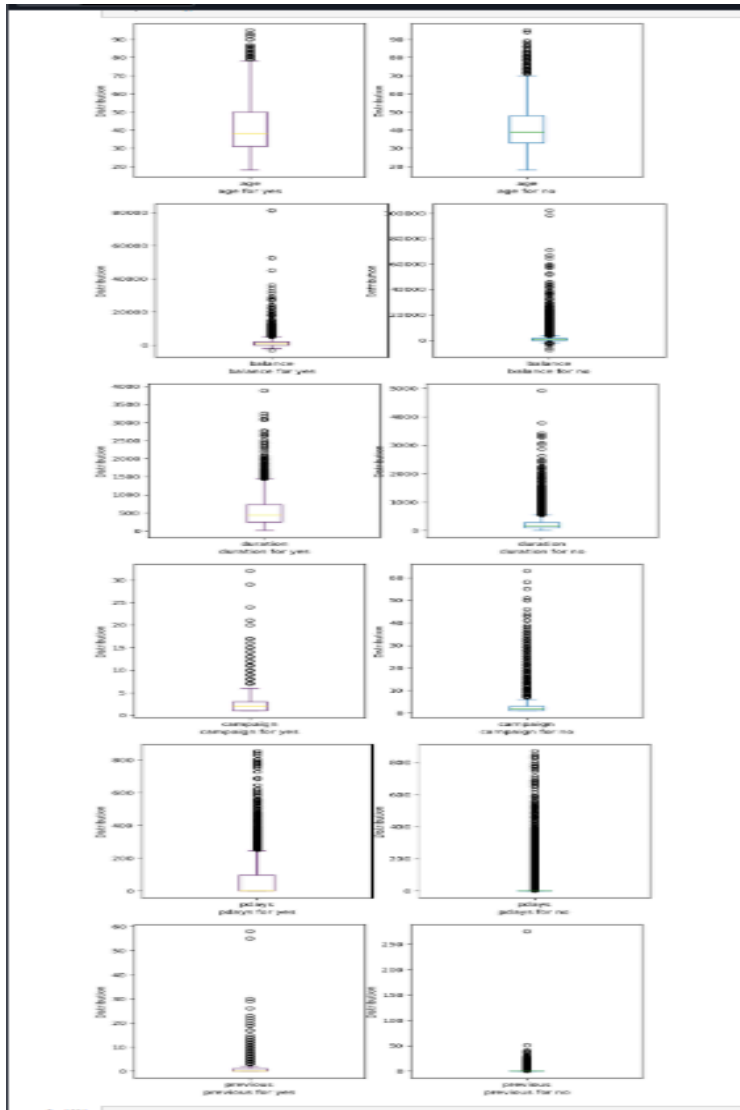
Personal Loan

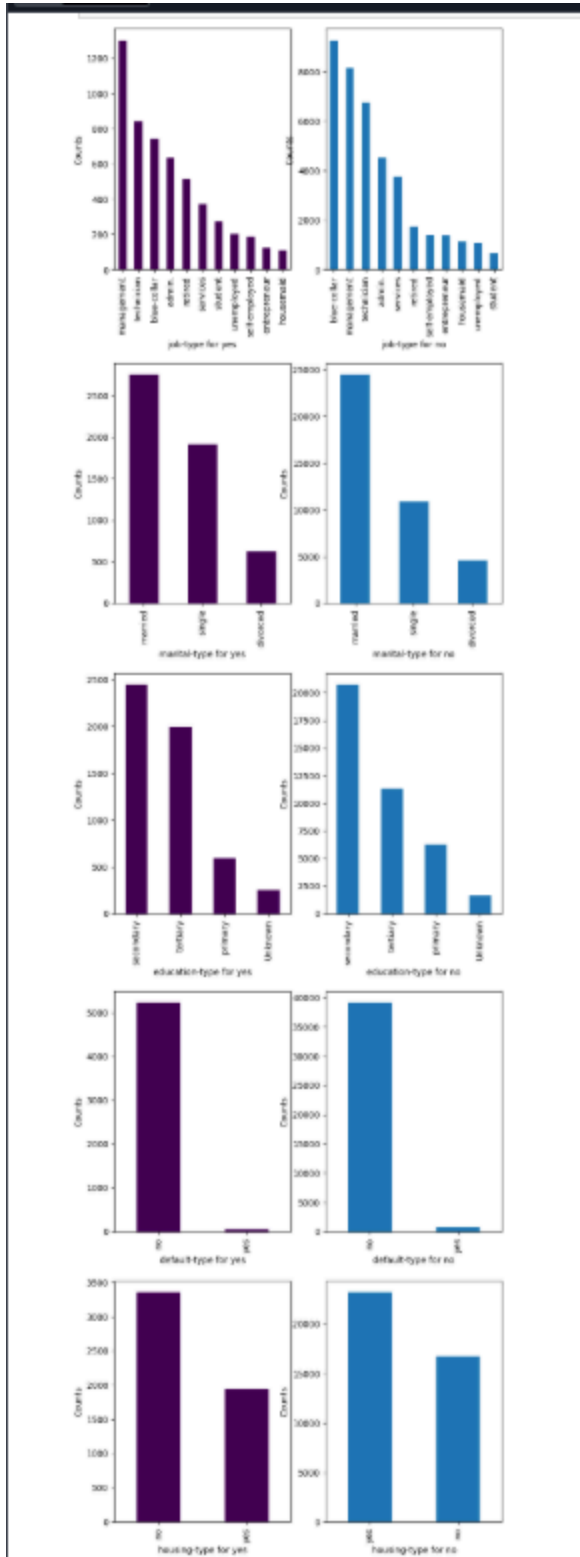
- Customers without personal loans have higher subscription rates.

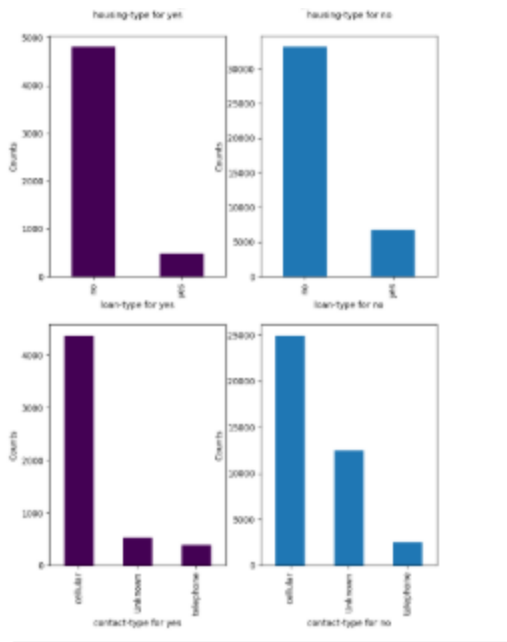
- Loan holders are less likely to subscribe, similar to housing loan behavior.

Contact Type

- Subscription proportions are similar across different contact methods.
- Contact type does not strongly influence customer conversion.







Output Variable Distribution:

The output variable is **imbalanced in distribution almost 88% has not subscribed** to term deposit **and only 12% have subscribed** to the term deposit.

Bivariate Analysis

Job and Marital Status

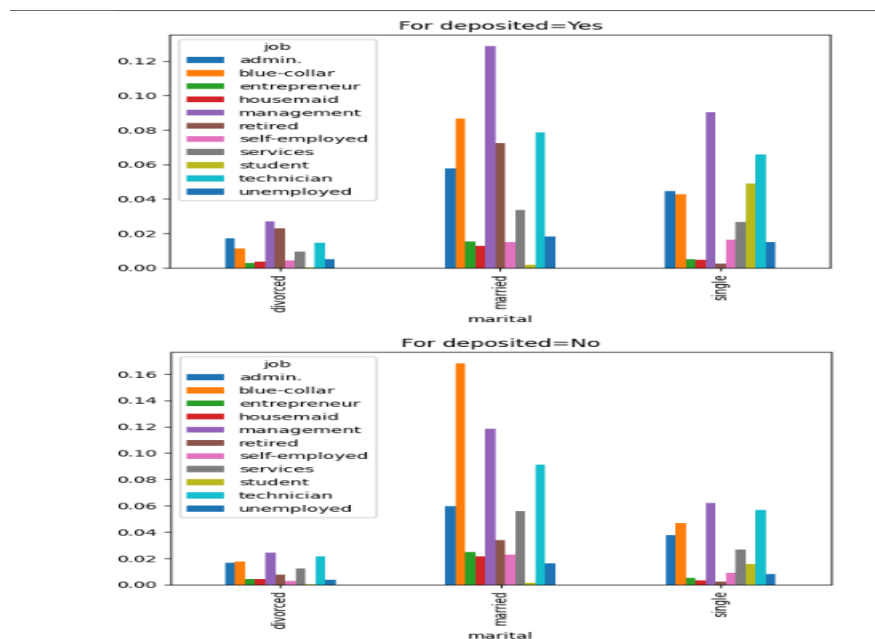
- Management professionals show the highest subscription rates, especially among married and single groups
- Blue-collar workers are the major chunk of non-subscribed group, particularly among married individuals
- Technicians and services roles show moderate behavior across both classes

Interpretation

Management + married → high probability

Blue-collar + married → low probability

Tree-based models (RF, GB) perform better

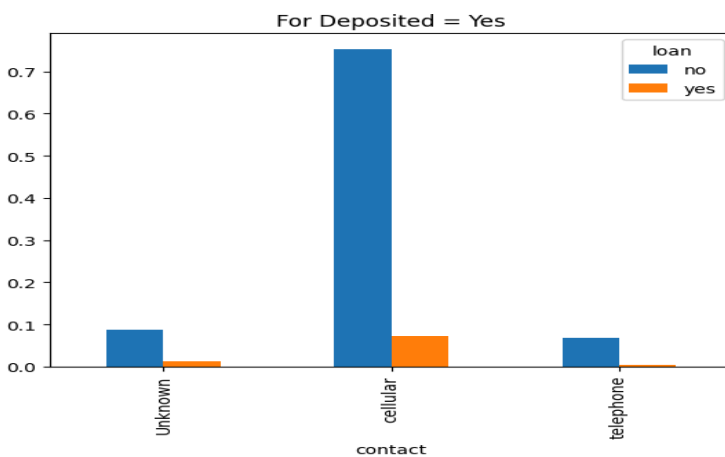


2.Contact type vs Loan

Cellular contact dominates conversions (both Yes and No).

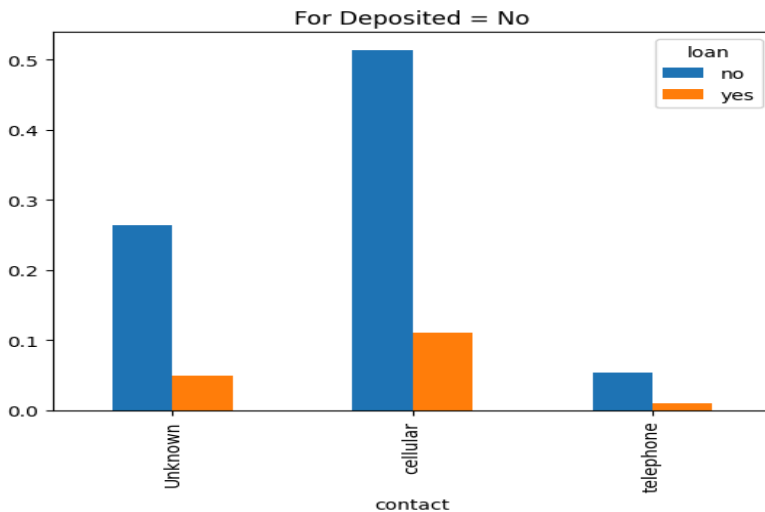
Customers without loans are more likely to subscribe, regardless of contact type.

"Unknown" contact shows weaker conversion patterns.



Interpretation

Contact alone is a weak predictor, but combined with loan it becomes stronger.



Default Status vs Loan

- Customers with **default = yes** almost never subscribe
- Majority of subscribers belong to **non-default category**

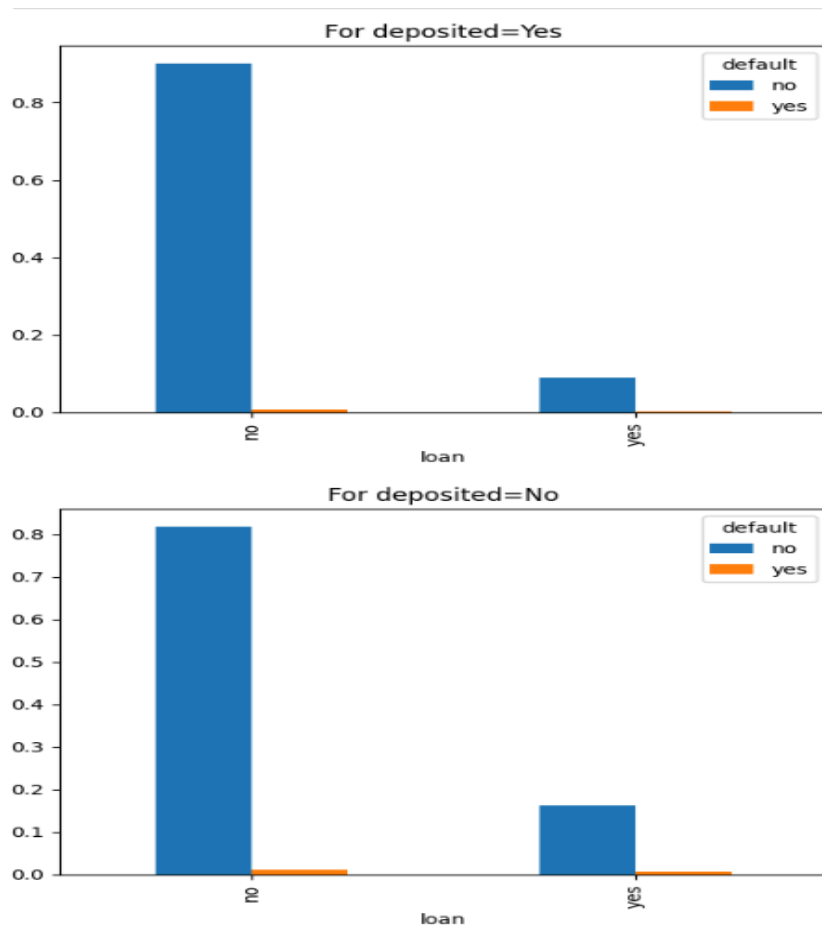
Interpretation

Default > Yes > Nearly 0 probability of Subscribing to Term Deposit

Conclusion

In ,Bivariate Analysis we found that **there are association between all features and their combined effect matters more than individual effects**

This indicates that **linear models(Logistic Regression)** will struggle for prediction rather than Non **Linear Models (Tree Based Models)** will work better as there is a conditional relationship between the features.



Machine Learning Modeling

Feature Engineering

- **Train Test Split : (Train 75% and 25% Test)** (With Shuffle True (to ensure randomness) and Stratify True (To maintain class distribution due to class imbalance)
- **Column Transformer Pipeline :** One Hot Encoder for Categorical columns ,with dropping First column

Model Selection

Logistic Regression

- A classification model built using Logistic Regression due to its Simplicity ,Interpretability ,Strong performance as a baseline
- Hyperparameter tuning using Randomized Search CV with K-fold Cross Validation


```
{
    'solver': 'saga',
    'penalty': 'l1',
    'max_iter': 1000,
    'C': 0.01,
    'class_weight': 'balanced'
}
```

Model Evaluation

	precision	recall	f1-score	support
0	0.92	0.75	0.83	9981
1	0.20	0.48	0.28	1322
accuracy			0.72	11303
macro avg	0.56	0.61	0.55	11303
weighted avg	0.83	0.72	0.76	11303

Accuracy: ~72%

Positive Class (Subscribers):

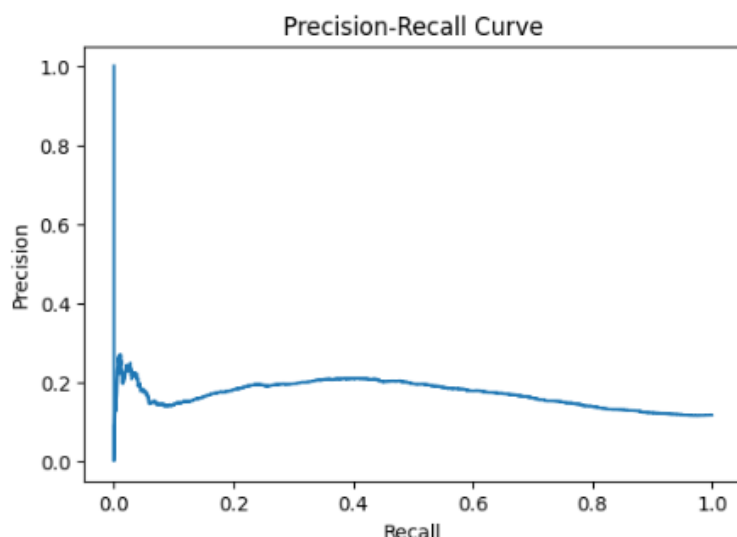
Recall: 0.48

Precision: 0.20

Selected optimal threshold = 0.49 based on F1-score maximization

The model is recall-focused means captures more actual subscribers and allows more false positives

If Missing a potential customer (False Negative) is more costly than contacting a non-interested one (False Positive)



Decision Tree Classifier

Model Selection

- A classification model built using Decision Tree due to its Conditional, Intuitive and Strong baseline for ensemble models.
- Hyperparameter tuning using Randomized Search CV with K-fold Cross Validation


```
{
        splitter': 'best',
        'min_samples_split': 20,
        'min_samples_leaf': 2,
        'max_features': None,
        'max_depth': 7,
        'criterion': 'gini',
        'class_weight': 'balanced',
        'ccp_alpha': 0.0001
      }
```

	precision	recall	f1-score	support
0	0.92	0.92	0.92	9981
1	0.40	0.38	0.39	1322
accuracy			0.86	11303
macro avg	0.66	0.65	0.65	11303
weighted avg	0.86	0.86	0.86	11303

Accuracy: ~86%

Positive Class (Subscribers):

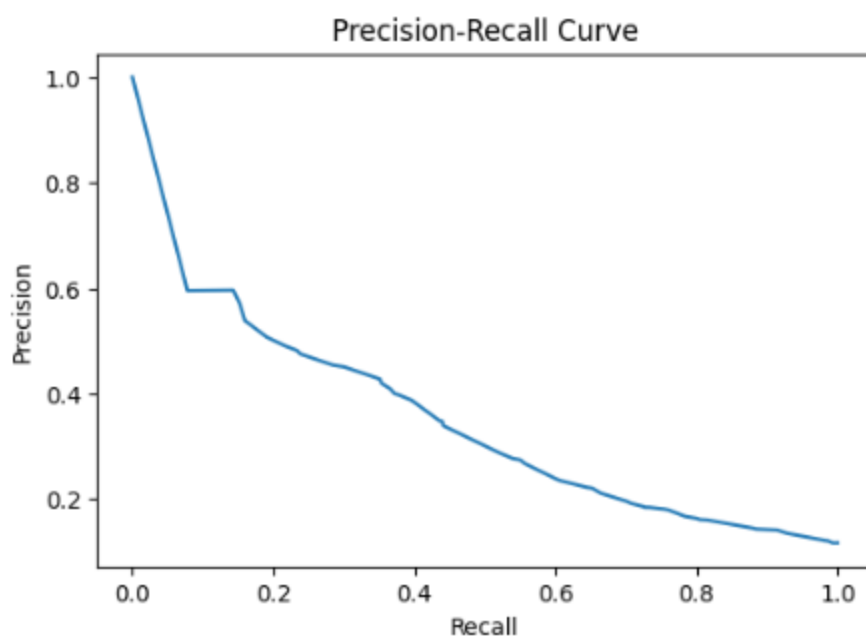
Precision: 0.40

Recall: 0.38

(F1 score based threshold with probability >0.66 classified as Positive Class)

The model shows **balanced precision and recall** for the positive class
It maintains a **controlled trade-off between False Positives and False Negatives**
This maintains a reasonable **control over unnecessary marketing outreach**

Decision Tree captures **non-linear relationships and feature interactions**
Provides better **interpretability** through **decision rules**
However, performance on the **minority class is moderate, indicating scope for improvement**



Random Forest Classifier: Model Selection

- A classification model built using Random Forest Classifier due to its Conditional, Intuitive and stable prediction using bagging.
- Hyperparameter tuning using Randomized Search CV with K-fold Cross Validation


```
{
    'oob_score': True,
    'n_estimators': 500,
    'min_samples_split': 10,
    'min_samples_leaf': 1,
    'min_impurity_decrease': 0.01,
    'max_features': 'sqrt',
    'max_depth': 15,
    'criterion': 'entropy',
    'class_weight': 'balanced',
    'ccp_alpha': 0.0
  }
```
-

	precision	recall	f1-score	support
0	0.91	0.93	0.92	9981
1	0.39	0.31	0.34	1322
accuracy			0.86	11303
macro avg	0.65	0.62	0.63	11303
weighted avg	0.85	0.86	0.85	11303

Accuracy: ~86%

Positive Class (Subscribers):

Precision: 0.39

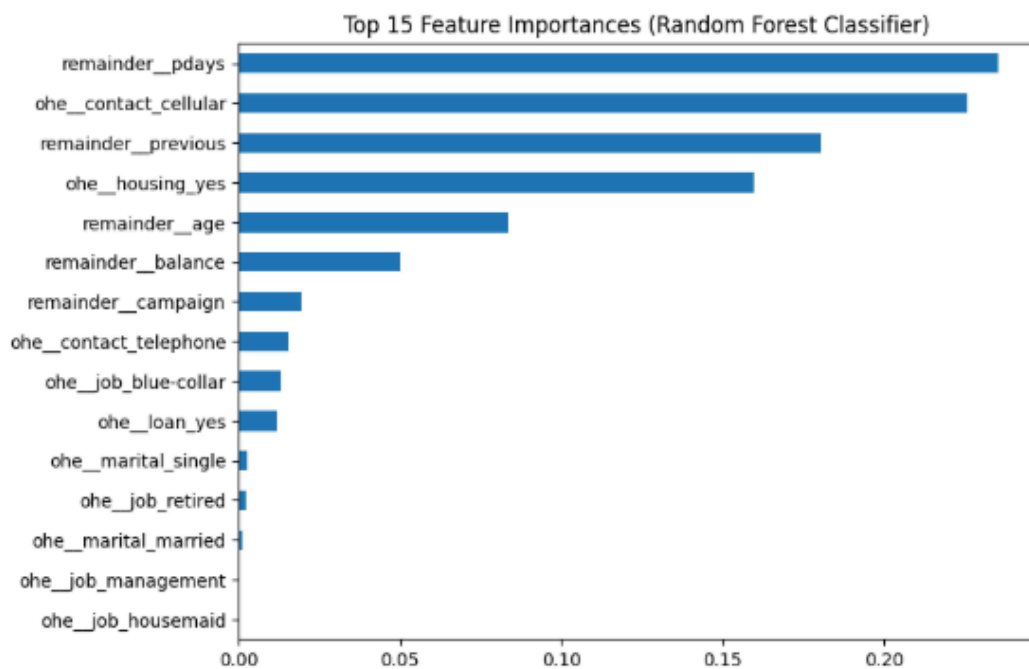
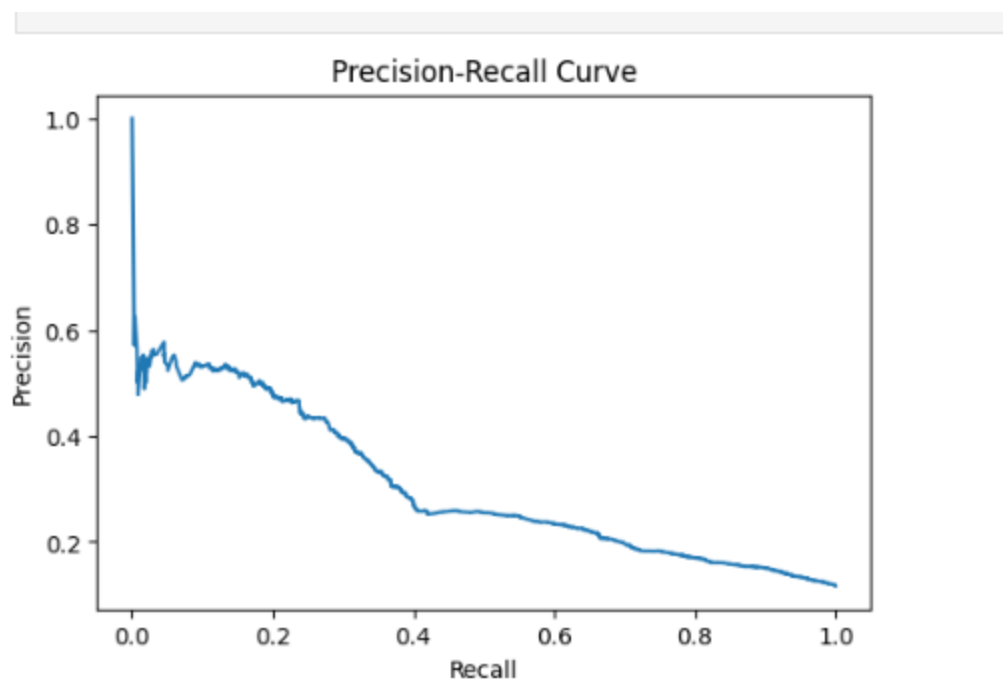
Recall: 0.31

Optimal threshold selected: 0.66 based on F1-score maximization

The model achieves **similar overall accuracy** to Decision Tree

However, it shows **lower recall for the positive class**, meaning: More actual subscribers are missed (higher False Negatives) **Precision remains comparable**, indicating controlled false positives.

Random Forest is more conservative in predicting positive class. It prioritizes certainty before classifying a customer as subscriber; also it **reduces overfitting** compared to Decision Tree and **captures complex feature interactions and non-linearity**. However, in this case, it leans toward the majority class, **reducing recall**.



Gradient Boosting Classifier: Model Selection

- A classification model built using Gradient Boosting Classifier due to its Conditional, Intuitive and stable prediction using boosting (learning from its errors).
- Hyperparameter tuning using Randomized Search CV with K-fold Cross Validation and used sample weights for training to handle class imbalance

```
{
  'n_estimators':500,
  'min_samples_split': 20,
  'min_samples_leaf': 1,
  'min_impurity_decrease': 0.01,
  'max_features':log2,
  'max_depth': 7,
  'criterion': 'friedman_mse',
  'ccp_alpha': 0.0001,
  'subsample':0.5,
  'validation_fraction':0.5
}
```

	precision	recall	f1-score	support
0	0.93	0.89	0.91	9981
1	0.36	0.47	0.41	1322
accuracy			0.84	11303
macro avg	0.64	0.68	0.66	11303
weighted avg	0.86	0.84	0.85	11303

Accuracy: ~84%

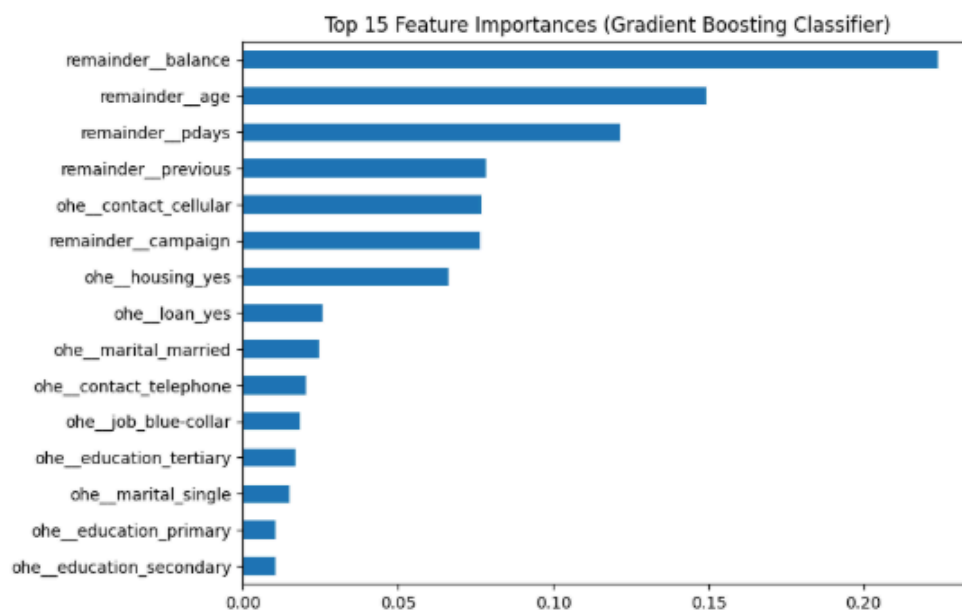
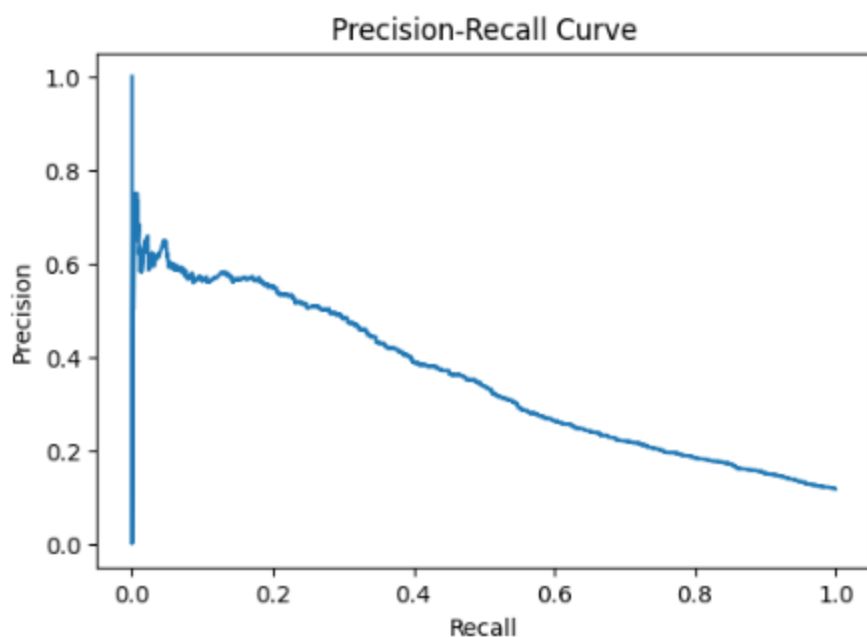
Positive Class (Subscribers): Precision: 0.36 ,Recall: 0.47

Optimal threshold selected: 0.64 ,Based on F1-score maximization

This results in a **better trade-off between marketing cost and missed opportunities**

Gradient Boosting helps in classifying Marketing data by handling **highly non-linear features** where Customer behavior depends on **combinations of features** .

Gradient Boosting improves **prediction for hard to classify clients and** focuses more on **misclassified samples** also it has **improved recall for subscribed class**



The Gradient Boosting model **emerged as the best-performing model**, achieving the **strongest balance between recall and precision for the subscriber class**. Its ability to **capture non-linear relationships, learn from misclassifications, and handle imbalanced data** makes it **highly suitable for predicting probability of subscribing to a term deposit plan**.